PRODUCT DEVELOPMENT · VISUAL PROJECT BLUEPRINT

Car Service Station Management SaaS

Bookings, job cards, parts, billing, staff pay and customer history for independent car service stations in Pakistan

APPROVED 2026-11-13 — development handoff (no code built yet)

VERSION	DOCUMENT DATE	SOURCE VERSIONS
1.0	**2026-11-13**	**PROJECT_CONCEPT_BRIEF.md v1.2 (APPROVED, incl. CR-002), PRD.md v1.1 (APPROVED, 61 active REQs, REQ-011 deprecated), ARCHITECTURE.md v1.0 (APPROVED), IMPLEMENTATION_PLAN.md v1.0 (APPROVED), TASKS.md v1.0 (APPROVED plan, 75 tasks, all NOT_STARTED), DECISIONS.md (DEC-001… DEC-032, CR-001 superseded, CR-002), Styling: neutral default colour — provisional design, no brand assets supplied**

Specification visualization only. Screen wireframes and diagrams do not represent finished software.

1 Project at a glance

Problem

Workshops run on paper job cards, registers and spreadsheets: parts used on a car are not always billed, stock and supplier dues are hard to track, customers cannot see progress, and owners lack daily cash and staff-performance figures.

Proposed solution

A multi-tenant SaaS with fully isolated data per station. Web dashboard for owner, manager and cashier; Android Staff app (mechanic work offline up to 24 h); Android Customer app for booking, estimate approval, tracking and invoices. Three plans: Basic PKR 2,500, Standard PKR 5,000, Premium PKR 8,000 per month; yearly = 10 months' price; 14-day full trial.

AUDIENCE

Independent car service stations in Pakistan (one branch each at launch), their staff and mechanics, their customers, and the platform owner as super admin. English UI, PKR, Pakistani formats.

FIRST RELEASE / MVP

- MOD-001…MOD-010, one branch per station, web + Android Staff app + Android Customer app (direct APK)
- Job cards with checklist (no photos), estimate approval and extra-work re-approval
- Parts by piece / litre / partial drum / kit, supplier purchases and ledger, low-stock alerts
- Invoices with optional sales tax (NTN/STRN), split payments, approved credit, refunds by credit note, daily cash closing
- Booking with owner-set bay capacity and no-show blocking (Standard+)
- WhatsApp: manual prefilled send on all plans; automatic from the station's own connected account on Standard+ (CR-002)
- Staff attendance, payroll with commission and advances (Premium)
- Approvals inbox, owner-only audit log, reports with Excel export

OUT OF SCOPE

- iOS apps (later)
- Multi-branch, stock transfers, branch-based plans (later)
- Urdu/RTL UI, FBR e-invoicing, Play Store listing, inspection photos (later)
- SMS for customer messages and any platform message quota (removed by CR-002; SMS only for login codes)
- In-app online payment by customers, marketplace/station search, white-label domains, general ledger accounting

2 People and permissions

Platform Super Admin

ACCESS AND PERMITTED ACTIONS

- Tenants, plans, trials, suspensions
- Verify bank/cash subscription payments
- Usage counts only — no station records without an owner-granted, time-limited support grant (REQ-005, REQ-012)

Station Owner

ACCESS AND PERMITTED ACTIONS

- Everything in the station: settings, prices, discount limit, tax
- Sees profit, purchase cost, audit log, payroll
- Approves discounts over limit, refunds, stock adjustments, credit release (REQ-057)
- Connects the station's WhatsApp account (REQ-061)

Manager

ACCESS AND PERMITTED ACTIONS

- Job cards, bookings, assignment, quality check
- Daily attendance (same day)
- Approval rights delegated by owner by default
- No profit, payroll or audit log

Cashier / Service Advisor

ACCESS AND PERMITTED ACTIONS

- Intake, estimates, invoices, payments
- Discount up to owner-set limit (default 5%)
- Requests approvals; sends manual WhatsApp (REQ-062)

Mechanic

ACCESS AND PERMITTED ACTIONS

- Only assigned jobs, no prices
- Checklist, stages up to quality check, parts used
- Works offline up to 24 h (REQ-023)

Customer

ACCESS AND PERMITTED ACTIONS

- One phone login across linked stations (REQ-047)
- Book free slots, approve estimates and extra work
- Track status, view invoices, dues and history

3 User journeys

Illustrative flows derived from the approved requirements.

J1 Booking to delivery (Standard)

1 Customer books free slot in app → 2 Arrives; job card with odometer, fuel, checklist → 3 Estimate approved in app → 4 Mechanic adds parts (offline OK) and syncs → 5 Manager QC pass → Ready message → 6 Invoice paid by JazzCash → 7 Delivered + next-service reminder → ✓ Stock reduced, invoice paid, customer informed via push and station WhatsApp

REQ-015 REQ-018 REQ-020 REQ-023 REQ-024 REQ-036 REQ-038 REQ-049 REQ-055

J2 Walk-in on Basic plan

1 Cashier creates walk-in job card → 2 Records phone approval → 3 Discount above limit needs owner login → 4 Invoice + Bluetooth receipt → 5 Manual WhatsApp send → ✓ Paid invoice; money control kept with the owner

REQ-018 REQ-020 REQ-037 REQ-045 REQ-062

J3 Stock in

1 Manager enters supplier bill → 2 Stock rises; average cost updated → 3 Supplier due rises (Standard+) → 4 Supplier payment reduces due → ✓ Correct stock value and supplier ledger

REQ-029 REQ-030

J4 Credit release

1 Customer can't pay full amount → 2 Cashier requests credit release → 3 Owner approves in Staff app → 4 Vehicle delivered; due in customer ledger → 5 Weekly dues reminder → ✓ Vehicle leaves only with approval; dues tracked

REQ-039 REQ-040 REQ-057

J5 Subscription lapse

1 Renewal date passes unpaid → 2 Days 1–7: full use + reminders → 3 Day 8: read-only → 4 Owner uploads bank receipt → 5 Super admin verifies → full access → ✓ Station keeps its data; access restored immediately on payment

REQ-008 REQ-009

J6 Month end payroll (Premium)

1 Daily attendance by manager → 2 Owner opens March payroll → 3 Salary by paid days + commission on paid labour − advances → 4 Owner finalises; sheet locks → ✓ Example from PRD: net Rs 28,000

REQ-051 REQ-052 REQ-053

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4 Module map

MOD-001

Tenant workspace, sign-up & access

Self sign-up, roles, strict station isolation, branding, support access.

FEATURES

- OTP-verified sign-up
- Role-based menus
- Row-level isolation
- Logo/NTN/STRN/tax settings

REQ-001 REQ-002 REQ-003 REQ-004 REQ-005

MOD-002

Subscription & platform admin

Plans, trial, payments, 7-day grace then read-only, tenant exit.

FEATURES

- 3 plans by features
- JazzCash/Easypaisa/card/bank/cash
- Grace + read-only
- Excel export, deletion after 90 days

REQ-006 REQ-007 REQ-008 REQ-009 REQ-010 REQ-012 REQ-013

MOD-003

Booking & scheduling

Owner-set bays and hours; customers auto-book free slots.

FEATURES

- Capacity per slot = bays
- Staff walk-in/phone bookings
- No-show after 30 min, block after N

REQ-014 REQ-015 REQ-016 REQ-017

MOD-004

Job cards & inspection

Intake to delivery with approvals, offline mechanic work, rework and warranty.

FEATURES

- Checklist (no photos)
- Estimate + re-approval
- Stages + QC
- Offline 24 h, last change wins

REQ-018 REQ-019 REQ-020 REQ-021 REQ-022 REQ-023 REQ-024 REQ-025 REQ-026 REQ-027

MOD-005

Parts, inventory & suppliers

Stock by movement ledger with weighted average cost.

FEATURES

- Piece/litre/drum/kit
- Supplier ledger (Standard+)
- Adjustments with approval
- Low/negative stock alerts
- Excel import

REQ-028 REQ-029 REQ-030 REQ-031 REQ-032 REQ-033 REQ-034

MOD-006

Billing, payments & customer credit

Invoices, discounts, tax, split payments, credit, refunds, day closing, printing.

FEATURES

- Labour service list
- Discount limit
- Credit only with approval
- Credit notes, no deletion
- Bluetooth / A4 / PDF

REQ-035 REQ-036 REQ-037 REQ-038 REQ-039 REQ-040 REQ-041 REQ-042 REQ-043 REQ-044 REQ-045

MOD-007

Customer & vehicle history

Records, customer app account with station links, history, reminders.

FEATURES

- Search by phone/plate
- Linked stations only
- Next-service reminders

REQ-046 REQ-047 REQ-048 REQ-049

MOD-008

Staff, attendance, payroll & commission

Premium: attendance, salary proration, commission on paid labour, advances.

FEATURES

- Daily attendance
- Commission % per mechanic
- Payroll finalise + lock
- Performance report

REQ-050 REQ-051 REQ-052 REQ-053 REQ-054

MOD-009

Notifications

Push to staff and customers; WhatsApp manual (all plans) or automatic from the station's own account (Standard+).

FEATURES

- Per-event toggles
- Messages-to-send list
- Meta Embedded Signup
- No customer SMS

REQ-055 REQ-056 REQ-061 REQ-062

MOD-010

Reports, approvals & audit log

Approvals inbox, append-only audit, reports and dashboard.

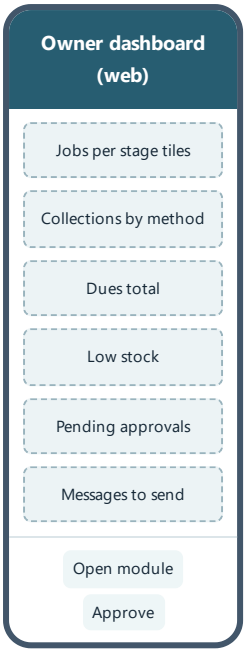
FEATURES

- 4 approval types
- Owner-only audit log
- Excel export
- Owner dashboard

REQ-057 REQ-058 REQ-059 REQ-060

5 Key screens — low fidelity

Each card is a schematic for discussion, not a designed or developed screen.



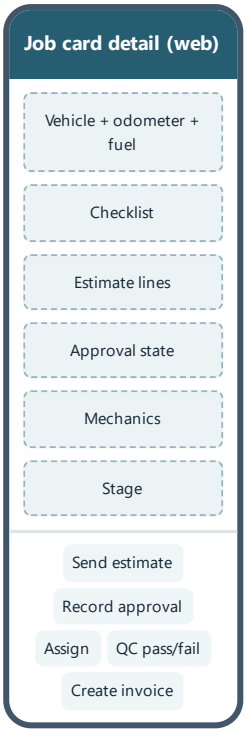
SCR-003

Owner dashboard (web)

Today at a glance

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-060 REQ-062



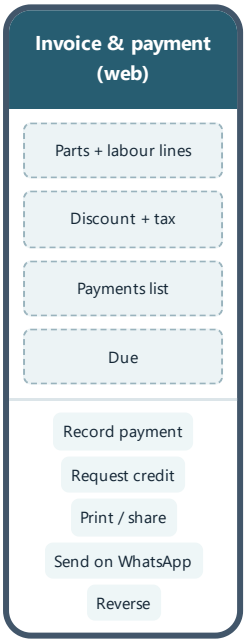
SCR-006

Job card detail (web)

Intake to quality check

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-018 REQ-019 REQ-020 REQ-021 REQ-022 REQ-024



SCR-007

Invoice & payment (web)

Bill and collect

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-036 REQ-037 REQ-038 REQ-039 REQ-045 REQ-062



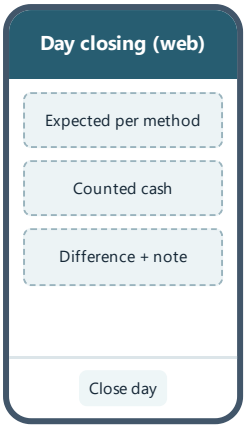
SCR-009

Inventory items (web)

Stock and prices

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-028 REQ-032 REQ-033 REQ-034



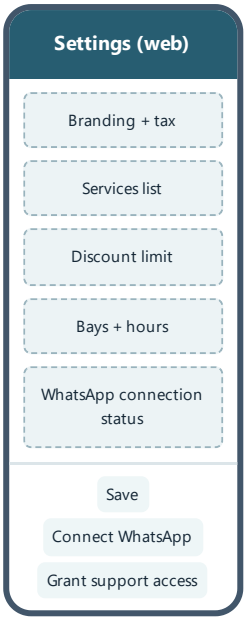
SCR-013

Day closing (web)

Match cash daily

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-044



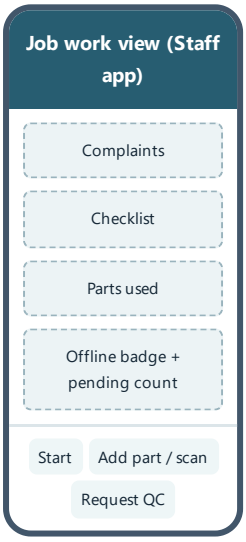
SCR-017

Settings (web)

Station rules

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-004 REQ-014 REQ-035 REQ-037 REQ-061 REQ-005



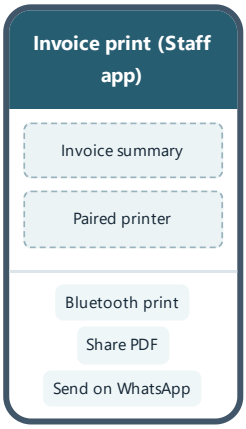
SCR-032

Job work view (Staff app)

Mechanic does the job, online or offline

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-021 REQ-023 REQ-031



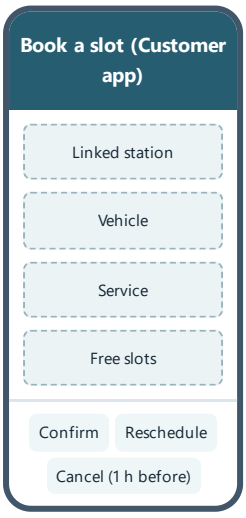
SCR-036

Invoice print (Staff app)

Receipt at the counter

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-045 REQ-062



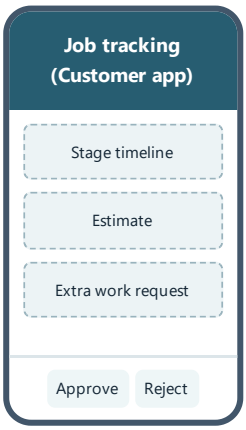
SCR-042

Book a slot (Customer app)

Self-booking

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-015 REQ-016 REQ-017



SCR-043

Job tracking (Customer app)

Follow and approve

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-020 REQ-022



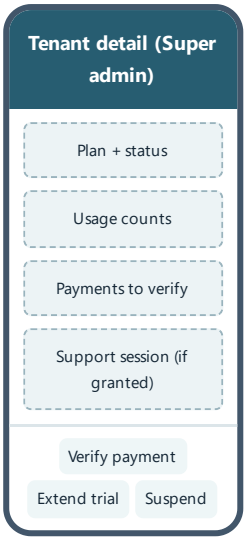
SCR-019

Subscription & billing (web)

Owner pays the platform

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-007 REQ-008 REQ-009 REQ-010



SCR-022

Tenant detail (Super admin)

Run the SaaS

LOW-FIDELITY WIREFRAME · NOT IMPLEMENTED

REQ-012 REQ-005 REQ-008

6 Proposed technical architecture

Clients (Flutter)

Web dashboard + super admin (Flutter web) · Staff app (Android 8+, local SQLite + change queue) · Customer app (Android 8+, online)

↓

Supabase — Mumbai ap-south-1

Auth (password; phone OTP via SMS hook) · Postgres with row-level security per station · Business rules in SQL functions (one transaction each) · Append-only stock, payment, ledger and audit tables · Edge Functions (webhooks, WhatsApp sending, exports) · Scheduler jobs (no-show, grace, reminders, deletion) · Storage (logos, receipts, PDFs, APKs)

↓

External services

Firebase Cloud Messaging (push) · WhatsApp Cloud API — each station's own account via Embedded Signup · Local SMS aggregator — login codes only · Payment aggregator — JazzCash, Easypaisa, cards

DOCUMENTED FLOWS

- Every money/stock/stage change goes through a named SQL function that checks role, plan, read-only state and validation, then writes ledgers and audit

TRUST BOUNDARIES

- Apps hold only the public URL and anon key; all access enforced by row-level security
- Provider keys and station WhatsApp tokens live only in Edge Function secrets

- Staff app offline: local write → change queue → sync_apply (idempotent by client_change_id) → last change wins, loser kept in history, manager alerted
- Event → message outbox → automatic send from the station's WhatsApp account, or the manual Messages-to-send list with a prefilled wa.me link
- Subscription: checkout → provider webhook (signature checked, status re-queried) → plan activated; bank/cash verified by super admin
- Backups: Supabase daily (7-day retention) + weekly off-site copy; quarterly restore drill
- Mechanic devices never receive prices or purchase costs; server re-validates every uploaded change
- Super admin sees station data only during an owner-granted, time-limited, read-only support grant

7 Data, content and knowledge

Station operations (jobs, invoices, payments, stock, staff)

Source: Staff entry on web and Staff app

Retention: While account active; deleted 90 days after closing

Customers and vehicles

Source: Staff entry, Excel import, customer app

Retention: Same as station data; customer account per phone links to stations

Audit log

Source: Database triggers (no edits or deletes)

Retention: Life of the account

Subscription payments

Source: Payment aggregator webhooks; super admin verification

Retention: Platform records

Backups

Source: Supabase daily + weekly off-site export

Retention: 7 days daily; 8 weeks weekly copies

8 Security and reliability constraints

- Cross-station isolation tests on every table in CI (TEST-003)
- Discount, credit, refund and stock rules enforced on the server, not only in the UI
- No deletion of money or stock records — reversal with reason only (REQ-042)
- Webhook signature checks + idempotency; OTP rate limit (5 per hour per phone, ARCHITECTURE §6)
- Signed APKs from our HTTPS domain; keystore kept offline with two copies
- Read-only mode on subscription lapse blocks writes in the database

- Targets: screens < 3 s on 4G, 99.5% monthly availability, 24 h offline for mechanics

9 Delivery roadmap — planned, not executed

Phase 0 — Setup & verification

Accounts, vendors and spikes verified

PLANNED TASKS

- TASK-001…TASK-009: repo + CI, Supabase Mumbai, hosting, Meta Tech Provider, SMS OTP, payment aggregator, own-sync spike, Flutter web spike, printer/scanner spike

REQ-003 REQ-008 REQ-023 REQ-045 REQ-061

Phase 1 — Foundation

Tenancy, roles, entitlements, audit

PLANNED TASKS

- TASK-010…TASK-019

REQ-001 REQ-002 REQ-003 REQ-004 REQ-005 REQ-006 REQ-058

Phase 2 — Inventory, approvals, customers

Stock, suppliers, approvals engine, customer records

PLANNED TASKS

- TASK-020…TASK-028

REQ-028 REQ-029 REQ-030 REQ-032 REQ-033 REQ-034 REQ-046 REQ-048 REQ-057

Phase 3 — Job cards & offline Staff app

Intake to QC, own sync engine

PLANNED TASKS

- TASK-029…TASK-037

REQ-018 REQ-019 REQ-020 REQ-021 REQ-022 REQ-023 REQ-024 REQ-025 REQ-026 REQ-027 REQ-031

Phase 4 — Billing & payments → M1 internal alpha

Basic-plan feature set complete

PLANNED TASKS

- TASK-038…TASK-046

REQ-035 REQ-036 REQ-037 REQ-038 REQ-039 REQ-040 REQ-041 REQ-042 REQ-043 REQ-044 REQ-045

Phase 5 — Subscription & super admin

Trial → grace → read-only → paid cycle

PLANNED TASKS

- TASK-047…TASK-052

REQ-007 REQ-008 REQ-009 REQ-010 REQ-012 REQ-013

Phase 6 — Booking & Customer app

Self-booking and customer approvals

PLANNED TASKS

- TASK-053…TASK-059

REQ-014 REQ-015 REQ-016 REQ-017 REQ-047 REQ-049

Phase 7 — Notifications & WhatsApp → M2 beta

Automatic + manual WhatsApp, staff push

PLANNED TASKS

- TASK-060…TASK-062

REQ-055 REQ-056 REQ-061 REQ-062

Phase 8 — Reports, dashboard, payroll

Premium feature set

PLANNED TASKS

- TASK-063…TASK-067

REQ-050 REQ-051 REQ-052 REQ-053 REQ-054 REQ-059 REQ-060

Phase 9 — Hardening, pilot → M3 launch

Security, load, E2E, device tests; 2–3 station pilot

PLANNED TASKS

- TASK-068…TASK-075

REQ-003 REQ-005 REQ-013 REQ-045

10 Acceptance checkpoints

- Blueprint content matches approved Brief v1.2, PRD v1.1, Architecture v1.0 and Plan v1.0 — no new features
- All 10 modules, 6 roles and 6 journeys shown with REQ IDs
- Screens are low-fidelity concept cards, not implemented UI
- Architecture shows trust boundaries and external services
- Roadmap links phases to TASK and REQ IDs
- Open items labelled; no implemented code implied (all tasks NOT_STARTED)
- PDF rendered on A4 and pages visually checked

11 Risks and decisions

Risks

- Own offline sync must be correct (ordering, duplicates, conflicts) — dedicated test suite, real-device spike (R5)
- Meta Tech Provider and template approval delays — manual WhatsApp covers the gap (R2)
- Stations may find Meta sign-up / payment method hard — guide, video and onboarding support (R1)
- Payment aggregator merchant approval time — bank/cash manual path still works (R3)
- Direct-APK installs and updates may put customers off — install guide, in-app updater (R6)
- Schedule ≈ 9 months to launch assumes 2 developers + part-time QA (ASSUMED)
- Running cost ≈ US$40–50/month fixed is an estimate; prices to be verified in Phase 0

Unresolved questions

- Team size — needed to re-estimate the ~9-month schedule
- Vendors for SMS (login codes) and payment aggregator — chosen in Phase 0 after verification
- Exact current prices (Supabase compute, SMS, aggregator fees) — TASK-002, TASK-005, TASK-006

12 Document provenance

Source artifact	Version	Status
PROJECT_CONCEPT_BRIEF.md	1.2	APPROVED 2026-11-12/13
PRD.md + TRACEABILITY_MATRIX.csv	1.1	APPROVED 2026-11-13
ARCHITECTURE.md	1.0	APPROVED 2026-11-13
IMPLEMENTATION_PLAN.md	1.0	APPROVED 2026-11-13
TASKS.md	1.0	APPROVED plan; all tasks NOT_STARTED

Source artifact	Version	Status
DECISIONS.md	DEC-001…DEC-032, CR-002	Current
blueprint_data.json / BLUEPRINT.md	1.0	APPROVED 2026-11-13